P R A A N®

Embedded Firmware Engineer Task

Note: The given problem statement does not relate to the existing, current and potential projects at Praan, but as an embedded firmware engineer, we would be keen to test your abilities with this hypothetical use case.

Tasks Overview:

- Microcontroller Selection
- Firmware
- Wiring Schematic
- MQTT Data Packet Format
- Power Budget (Optional)
- Hardware Requirement Sheet (Optional)

Problem Statement:

Design the architecture for a device based on any microcontroller with WIFI and BLE for the below listed functionality

1. Microcontroller Selection

- Select a microcontroller (ESPRESSIF/NRF) that supports the listed peripherals.
- Assign relevant pins for each peripheral.
- Document your selection and pin assignments.

2. Firmware Flowchart

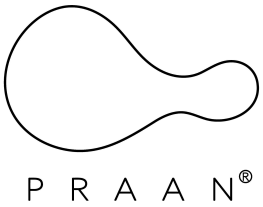
- Create a detailed firmware flowchart to cover the following modes of operation:
- The device to sample data continuously at 1Hz and publish to MQTT server at 0.5Hz: Please account for network loss and data backup due to same

3. Firmware

- Please write the firmware for the selected microcontroller in ESP IDF and share the relevant files to us.
- You can refer to the individual module datasheets, libraries and flag any issues observed, take necessary assumptions and recommend alternatives for optimization of hardware.

3. Wiring Schematic

- Create a wiring schematic for the controller board focusing on the MCU and its peripherals. Exclude power distribution, connectors, and motor driver schematics. If possible ensure hardware optimization with respect to the number of components.



4. MQTT Data Packet Format

Define MQTT data packet formats, including:

- Topics & Data packet structure
- Message QoS

5. Power Budget (optional)

- Calculate the power budget for the moving straight state of operation.
- Include voltage outputs and a reasonable factor of safety (FOS)

6. Hardware Requirement Sheet

- Document component selections, drivers, etc
- Include block diagrams as needed
- Specify the power requirements to be sent to a vendor for an off-the-shelf power distribution system for the rover

Peripherals to be Used:

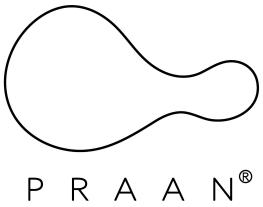
- BME280 (environmental sensor)
- SHT30 (humidity and temperature sensor)
- PMS5003 PM sensor
- DS3231 RTC module
- Data storage (1GB)

Assumptions:

- Assume a suitable power distribution system will be designed based on provided specifications (Not to be included in the schematic)

Deliverables:

1) Microcontroller Selection Document:
    - MCU selection with justification
    - Pin assignment for each peripheral
2) Firmware:
    - Detailed flowchart covering all modes of operation.
    - Firmware executable
3) Wiring Schematic:
    - Schematic for MCU and peripherals (You may work only on the block relevant to MCU and its peripherals. Excluding the following: power distribution, connector,OC/OV protection from the schematic, motor driver schematic)
4) MQTT Data Packet Format Document:
    - Topic names, data packet structure and message QoS.
5) Power Budget Document (optional)



- Voltage outputs and power calculations consider moving straight mode as the highest power consuming mode

6) Hardware Requirement Sheet
   - Detailed component selection, block diagrams, and power requirements

Best of luck!